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Programming In C-HDOC DAY_12

Problem

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits. **E.g.** Number: 12345, Sum = 6, Product = 5
- Count how many digits are lesser than, equal, and greater than a given digit `k`. **E.g.** Number: 12345, k=3, Count_lesser=2, Count_equal=1, Count_greater=2

ALGORITHM

- Start**
- Declare integer variables c and num.
- Ask the user to enter a positive integer num.
- Ask the user to enter a choice:
 - Choice 1:** Find the sum and product of the first and last digits.
 - Choice 2:** Count digits that are less than, greater than, or equal to a given number k.
- Read the choice c.
- Use a switch statement based on c.

Case 1: First and Last Digit

- Find the last digit using:
 $ld = num \% 10$
- Find the first digit using:
 $fd = num / 10^{(\text{number of digits} - 1)}$
- Calculate:
 - $sum = fd + ld$
 - $product = fd \times ld$
- Display the sum and product.
- Exit Case 1.

Case 2: Compare Digits with k

- Ask the user to enter a positive integer k.
- Initialize lt = 0, gt = 0, and eq = 0.

14. Repeat while num > 0:

- Extract the last digit: digit = num % 10
- If digit < k, increment lt.
- Else if digit > k, increment gt.
- Otherwise, increment eq.
- Remove the last digit: num = num / 10.

15. Display the number of digits less than, greater than, and equal to k.

16. Exit Case 2.

Default Case

17. If the choice is not 1 or 2, display "**Invalid choice**".

18. **Stop**.

C PROGRAM CODE

```
#include <stdio.h>
#include <math.h>

int main()
{
    int c;
    int num;

    printf("enter a positive integer number\n");
    scanf("%d", &num);

    printf("choice one is to find the sum and product of first and last digit of number entered\n");
    printf("choice 2 is to find how many digits of nos are less than or equal to or greater than the entered number k\n");
    printf("enter case number\n");
    scanf("%d", &c);
    switch (c)
    {
        case 1:
        {
            int fd,ld,sum,product,count=0;
            ld=num%10;
            fd=num/pow(10,((int)log10(num)));
            sum=fd+ld;
            product=fd*ld;
```

HDOC_DAY12 EXPERIMENT

```
        printf("the sum is %d and product is%d\n",sum,product);
        break;
    }
    case 2:
    {
        int k;
        printf("enter a positive integer number k\n");
        scanf("%d", &k);
        int lt=0,gt=0,eq=0;
        while(num>0)
        {
            int digit=num%10;
            if(digit<k)
                lt++;
            else if(digit>k)
                gt++;
            else
                eq++;
            num/=10;
        }
        printf("numbers less than k: %d\n", lt);
        printf("numbers greater than k: %d\n", gt);
        printf("numbers equal to k: %d\n", eq);
        break;
    }
    default:
        printf("invalid choice\n");
        break;

}
return 0;
}
```

TEST CASES

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	enter a positive integer number 112233 choice one is to find the sum and product of first and last digit of number entered choice 2 is to find how many digits of nos are less than or equal to or greater than the entered number k enter case number 1	the sum is 4 and product is 3	the sum is 4 and product is 3	PASS
2	enter a positive integer number 112233 choice one is to find the sum and product of first and last digit of number entered choice 2 is to find how many digits of nos are less than or equal to or greater than the entered number k enter case number 2 enter a positive integer number k 2	numbers less than k: 2 numbers greater than k: 2 numbers equal to k: 2	numbers less than k: 2 numbers greater than k: 2 numbers equal to k: 2	PASS